

1. What If There Were No Operating System? Imagine a computer without an operating system. Identify the problems a user would face and investigate how OS features solve them. Demonstrate at least three features using Linux and present a **“Without OS vs With OS”** comparison.

OS is an interface between the systems software and hardware. It performs multiple functionalities such as-

- *Manages the CPU, Memory and the files

- *It acts as an interface between the user and the hardware.

- *It performs the core functions like process, Memory and file management

Without an OS, it would be very difficult to carry out all these functionalities

CPU Management- There would no system to decide which program would get the CPU at what time and each program would have to manage the CPU by itself which would be very time consuming. The OS manages the CPU and will allocate the CPU time to different processes according to different criterias which will allow the processes to run efficiently.

Memory Management- Programs would directly have to manage the computer's memory which would lead to issues between different programs and inefficient usage of RAM. The OS manages the main memory and allocates memory to programs when they need it and releases it when its not required.

File Management- The user needs to directly interact with the storage device and have to create, access, modify and delete files by themselves. OS provides a file system that allows used to easily create, access, modify and delete files.

User Hardware interaction- The user would not be having any interface to communicate with the hardware. All the applications will have to communicate with the hardware directly. The OS will act as an intermediate and the user does not have to directly interact with the hardware.

The following are 3 features that can be demonstrated using linux-

1. CPU Management-

Linux manages CPU controlling and scheduling process.

Ex- ps

```
└─(kali㉿kali)-[~]
```

```
└─$ ps
```

```
PID TTY      TIME CMD
```

```
16368 pts/0    00:00:00 zsh
```

16699 pts/0 00:00:00 ps

2.Memory Management-

Linux manages the system's RAM and keeps track of how much memory is being used and how much is available.

Ex- free -h

```
└─(kali㉿kali)-[~]
```

```
└─$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	1.9Gi	802Mi	587Mi	9.8Mi	733Mi	1.1Gi
Swap:	953Mi	0B	953Mi			

3.File Management-

Linux provides a file system that allows users to create, access, modify, and delete files and directories.

Ex-mkdir

```
└─(kali㉿kali)-[~]
```

```
└─$ mkdir lab1
```

```
└─(kali㉿kali)-[~]
```

```
└─$ cd lab1
```

```
└─(kali㉿kali)-[~/lab1]
```

```
└─$ touch lab1.txt
```

```
└─(kali㉿kali)-[~/lab1]
```

└─\$ ls

lab1.txt

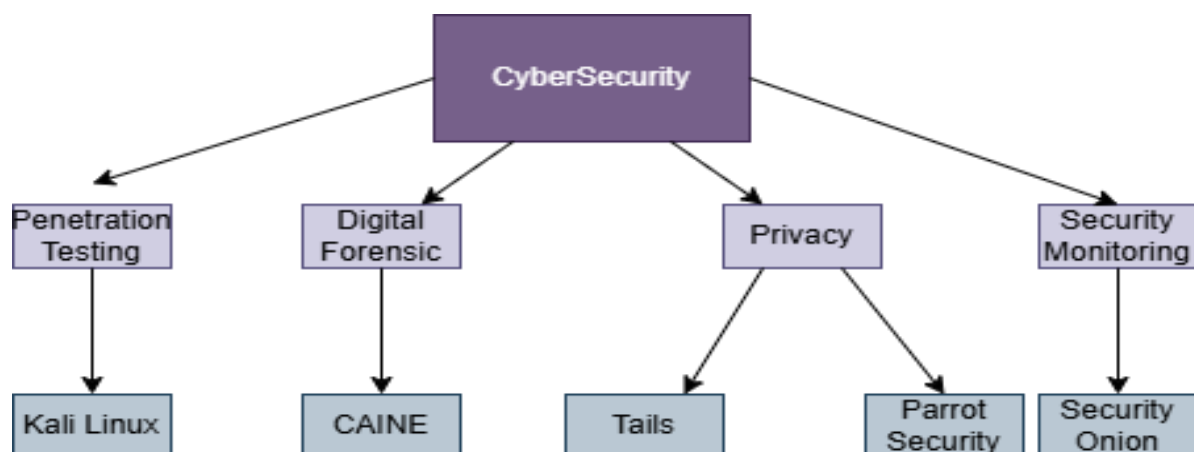
Functions	With OS	Without OS
CPU Management	Programs would have to manage the CPU by itself	Linux manages processes and allocates the CPU time
Memory Management	Programs would directly have to manage the computer's memory	Linux allocates and manages the memory
File Management	The user needs to directly interact with the storage device	Linux provides a file system and even the commands to manage the files
Hardware Interaction	The user would not be having any interface to communicate with the hardware	Linux provides a terminal/GUI as an interface

2. Imagine your team is setting up a cybersecurity laboratory. Explore at least five Linux distributions relevant to cybersecurity and recommend suitable distributions for different roles such as penetration testing, digital forensics, privacy, and security monitoring. Present your findings creatively as a comparison chart, poster, infographic, or “OS selection guide.”

Cybersecurity roles	Linux distributions	Advantages	Disadvantages	Features
Penetration testing	Kali Linux	It has a large collection of preinstalled penetration testing tools	It can be complex for beginners as some of the tools require advanced knowledge	Nmap, Metasploit, Wireshark
Digital Forensics	CAINE	It provides many forensic tools in one place and it also helps to preserve and analyze digital evidence	It mainly focuses on forensic work so it requires knowledge of forensic techniques	Disk imaging, evidence analysis, file recovery, forensic investigation
Privacy	Parrot security	It is lightweight and includes security tools. It can	It is not as specialized for anonymity as much as Tails	Security tools, encryption, secure browsing

		be used for everyday tasks as well as security testing		
Privacy	Tails	It is designed specifically for privacy and anonymity. It can run from a USB drive and routes internet traffic through tor	Some applications have limited functionality	Tor, Privacy focused, encrypted storage
Security Monitoring	Security Onion	It is designed specifically for security monitoring. It supports intrusion detection and threat hunting	The setup can be complex	IDS/NSM network monitoring, log analysis, threat hunting

OS selection guide



3. Exit status:

Execute at least five Linux commands that produce both successful and unsuccessful results. Investigate their exit status using ``$?``.

Create a simple visual/table showing *Command → Result Exit Status → Meaning*.

What can a script learn from an exit status?

A script can understand if the command has executed successfully using an exit status. If the exit status is 0 then the command was executed successfully, if its anything other than 0 then the command has failed or encountered some error.